← Back **Life and Icy Bodies**
Graded Quiz • 30 min

Due Mar 20, 11:59 PM CET

Congratulations! You passed!

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1. Which are the four 'Galilean' moons of Jupiter?

1 / 1 point

- [x] Callisto
  - Correct
    This is correct. It was one of the four moons first observed by Galileo.
- [ ] Charon
- [x] Europa
  - Correct
    This is correct. It was one of the four moons first observed by Galileo.
- [x] Ganymede
  - Correct
    This is correct. It was one of the four moons first observed by Galileo.
- [x] Io
  - Correct
    This is correct. It was one of the four moons first observed by Galileo.
- [ ] Titan
- [ ] Triton

2. Europa is said to be 'tidally locked' to Jupiter. What does this mean?

1 / 1 point

- ( ) That the tides are always the same
- ( ) That the moon swings rapidly from one side to another
- (•) That one face always points towards Jupiter
- ( ) That tides change very rapidly

Correct
This is correct. A tidally locked moon keeps one face towards its planet

3. The surface of Europa appears to be very young. What is one line of evidence for this?

1 / 1 point

- (•) There are very few impact craters
- ( ) It is white
- ( ) There are no ancient rocky mountain ranges
- ( ) It looks new

Correct
This is correct. The low numbers of craters suggests the surface is being reformed.

4. Magnetic measurements and other studies suggest that the icy crust of Europa directly covers what?

1 / 1 point

- ( ) A rocky core
- ( ) An ocean of liquid neon
- (•) An ocean of water
- ( ) A layer of solid organics

Correct
This is correct

5. Which 1970s spacecraft prompted discussions on the possibility of a subsurface ocean in Europa?

1 / 1 point

- ( ) The Viking spacecraft
- (•) The Voyager 1 and 2 craft
- ( ) The New Horizons spacecraft
- ( ) The Venera craft

Correct
This is correct. In 1979 they sent back detailed images of the surface that suggested a young age and began discussions about whether the moon contained water

6. Fractures on the surface of Europa are thought to be caused by what process?

1 / 1 point

- ( ) Ancient cracks from the formation of Europa
- ( ) Boiling water melting the ice
- ( ) Ice breaking apart under Europa's gravity
- (•) The tidal forces of Jupiter pulling on the moon and causing the fractures

Correct
This is correct

7. Why has Europa become a focus in the search for life beyond the Earth?

1 / 1 point

- ( ) It is full of organics
- ( ) It is far out in the Solar System
- (•) The presence of a large body of liquid water makes it a potentially favourable location for life
- ( ) It is a small planetary body

Correct
This is correct

8. Enceladus is a moon of Saturn and was discovered in 1789 by which astronomer?

1 / 1 point

- ( ) Tycho Brahe
- ( ) Galileo Galilei
- ( ) Christian Huygens
- (•) William Herschel

Correct
This is correct

9. The Voyager spacecraft observed what features on the surface of Enceladus that suggested a young ice surface?

1 / 1 point

- (•) A reflective surface with few craters
- ( ) Giant icebergs
- ( ) Melting water on the surface
- ( ) Frozen lakes and ponds

Correct
This is correct

10. The Cassini spacecraft observed plumes of material emanating from Enceladus. From where were they observed to come?

1 / 1 point

- ( ) From the surface at the equator
- ( ) From the equatorial region
- ( ) From the north polar region
- (•) From the south polar region

Correct
This is correct. The plumes come from 'Tiger stripes' in the south polar region

11. What component of the plumes of Encaladus has attracted the interest of astrobiologists?

1 / 1 point

- ( ) Sulfur
- ( ) Amino acids
- ( ) Bacteria
- (•) Organic material

Correct
The Cassini spacecraft detected complex organics in the plumes of Enceladus

12. Which of the following, as well as complex organics, have also been detected in the plumes of Enceladus?

1 / 1 point

- [x] Methane
  - Correct
    Methane has been detected
- [x] Propane
  - Correct
    Propare has been detected
- [x] Carbon dioxide
  - Correct
    Carbon Dioxide has been detected
- [x] Water
  - Correct
    Water has been detected

13. The Tiger Stripes from which the plumes of Enceladus emanate are thought to be about how old?

1 / 1 point

- ( ) 5 million years old
- ( ) 50 million years old
- (•) Less than 500,000 years old

Correct
This is correct. Geologically they are quite young

14. Which moon of Neptune is thought to have an icy crust and may once have hosted an ocean?

1 / 1 point

- ( ) Titan
- ( ) Titanic
- (•) Triton
- ( ) Titania

Correct
This is correct

15. In addition to Europa, other moons of Jupiter may host liquid water oceans. Which of the following is a possible candidate?

1 / 1 point

- [x] Ganymede
  - Correct
    This is correct
- [ ] Thebe
- [ ] Io
- [ ] Amalthea

16. Titan, a moon of Saturn has lakes and rivers containing which liquid substance?

1 / 1 point

- ( ) Carbon dioxide
- ( ) Sulfur
- (•) Methane
- ( ) Water

Correct
This is correct. There may also be other organics within the methane, but at Titan's low temperatures, methane is liquid

17. The surface features of Titan seem to move around suggesting what?

1 / 1 point

- (•) That Titan may host a subsurface ocean
- ( ) That Titan is all made of liquid
- ( ) That Titan is collapsing

Correct
This is correct

18. The organic rich surface and atmosphere of Titan has been compared to what?

1 / 1 point

- ( ) The formation of organics in the interstellar medium
- ( ) The fate of organic compounds in stars
- ( ) Life in deep sea vents
- (•) Early organic reactions leading to the origin of life on the Earth

Correct
This is correct. The formation of organic compounds might give us clues as to how complex organic chemistry occurred on the early earth

19. To date, what is the furthest body from the Sun in or near our Solar System that has been proposed as a location for a subsurface water body?

1 / 1 point

- (•) Charon, the moon of Pluto
- ( ) Europa, a moon of Jupiter
- ( ) Triton, a moon of Neptune
- ( ) Titan, a moon of Saturn

Correct
This is correct. This has been proposed, although we still don't know about the composition of Charon.